Classification Accuracy Assessment

Sources of Classification Error

- Errors may be caused by
  - Misidentification of parcels / Inaccurate selection of Training samples
  - Excessive generalization
  - Variations in detail of interpretation
  - Analyst's experience

Accuracy Assessments

- To determine the quality of the information derived from RS data through the classification process
- Purpose:
  - To validate the classification results , may be used for decision-making processes
  - End user must have a clear view of the quality of information before using it for any practical application

Accuracy Assessments

- Compare two sources of information
  - Pixels or polygons from Classified RS image
  - Ground reference information (test data, a standard)
- If the Image classification results corresponds closely with test data, the processing is “Accurate” or of high accuracy
- In statistical terms , high accuracy means a low bias between the estimated/ classified and reference values

Accuracy Assessments

- For accuracy assessment, the classified image and the reference data must be co-registered, such that both are at comparable level

Accuracy Assessments

- Types:
  - Qualitative
  - Quantitative
- Qualitative assessments:
  - By comparing the classified image with some reference maps, high resolution aerial images or the ground truth data (generally visual)

Assessment Types

- Quantitative assessments
  - To check the agreement between the classified and the reference data, to identify different types of classification errors

Quantitative Assessments

- As the quantitative accuracy assessment relies upon the reference data, it is imperative that the reference data be highly accurate
- It is better to get fewer, but more accurate reference data which are truly representative of the classes under assessment

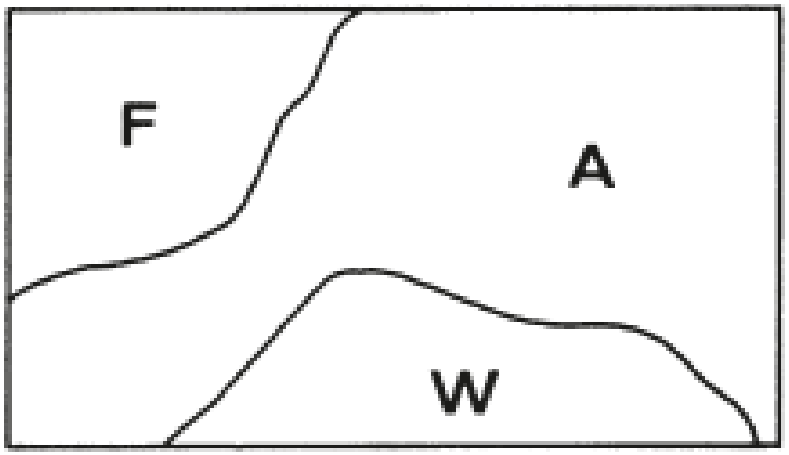
Quantitative Assessments

- Quantitative accuracy assessments can be performed as:
  - Non-site-specific assessments
  - Site-specific assessments
  - Error Matrix

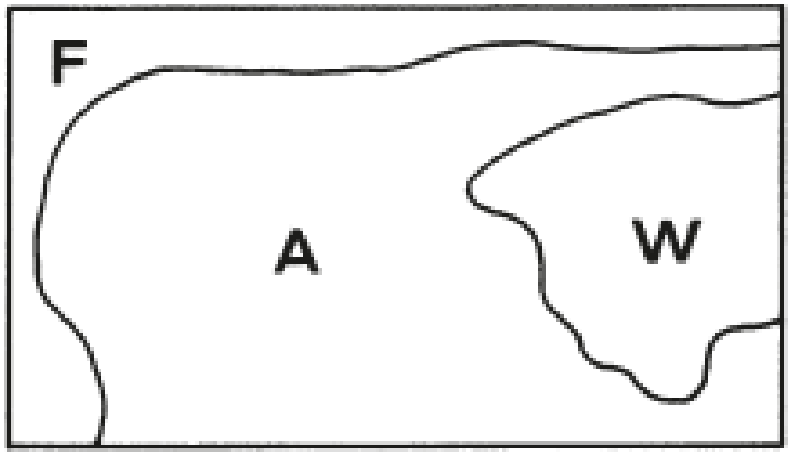
Quantitative Assessments

- Non-site-specific assessments
  - Does not consider agreement between the two maps at specific locations, but only the overall figures for the two maps (areas-based comparison)
  - Even though there may be close agreement between estimated class areas, the classification even then be inaccurate in terms of locational error, (no information about the locational accuracy)

Non-site-specific assessments



F = 20% W = 18% A = 62%

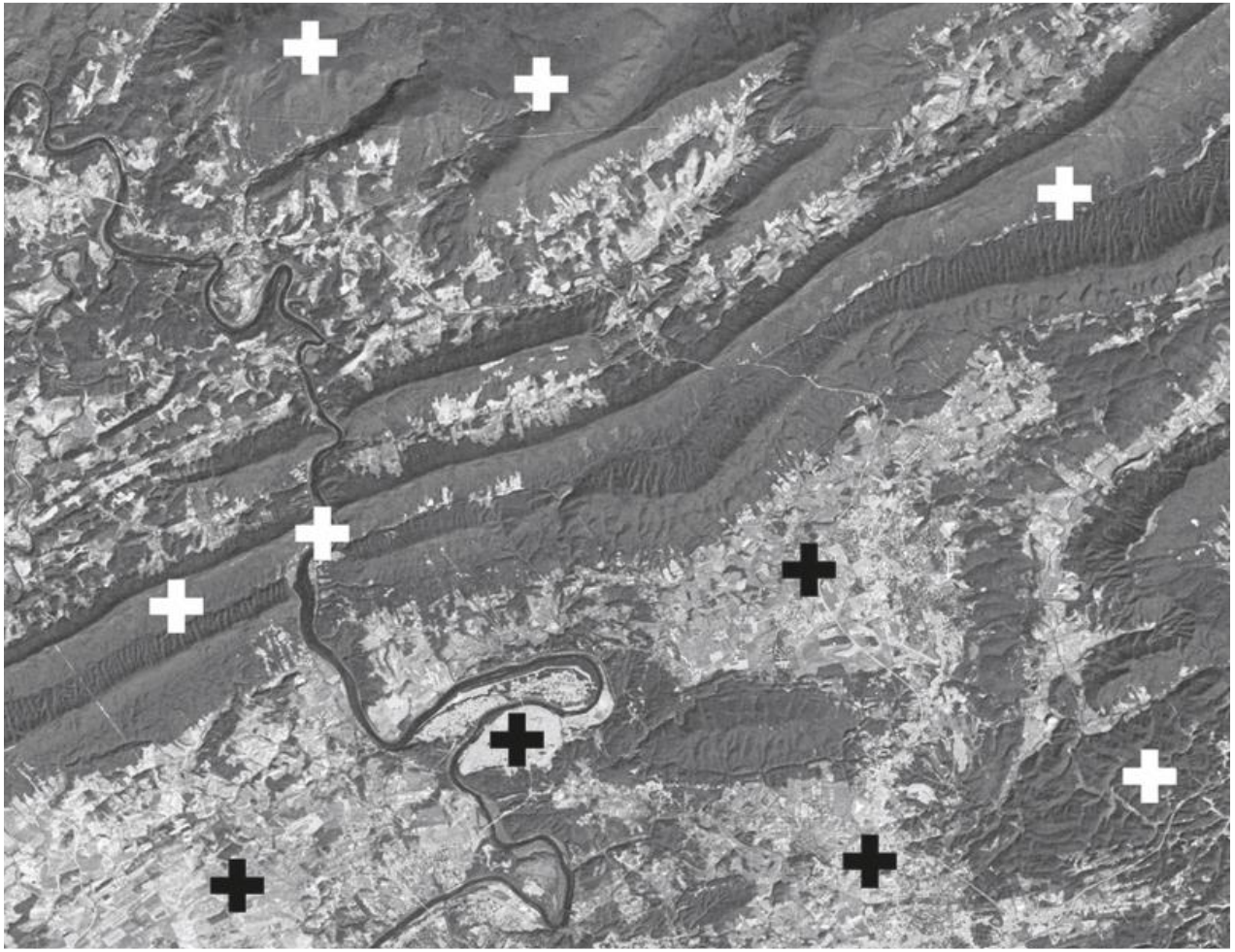


F = 20% W = 18% A = 62%

Quantitative Assessments

- Site-specific assessments
  - Assessment of the agreement between the classified and reference maps at specific locations

Site-specific Assessments

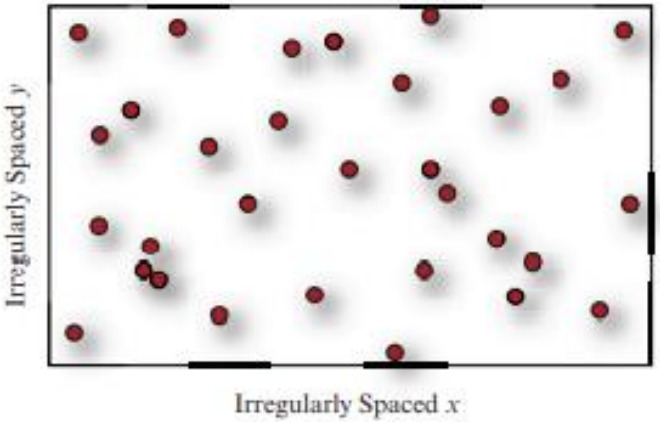


Reference data (shown here as black or white crosses) with known locations are used to assess map accuracy

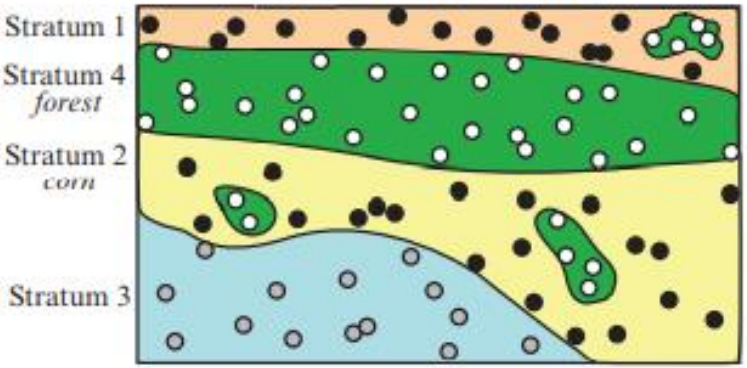
Sampling

- **What sampling scheme should be used?**
- **How many samples should be collected?**
- **Random Samples: is best with a large enough sample size,**
  - **However, as sample size decreases, categories that are less prevalent on the landscape can be missed using a simple random sample**
- **Stratified random sample: based on class categories**
- **“Sufficient” sample size? ----50 –100 samples per category as a general guideline**

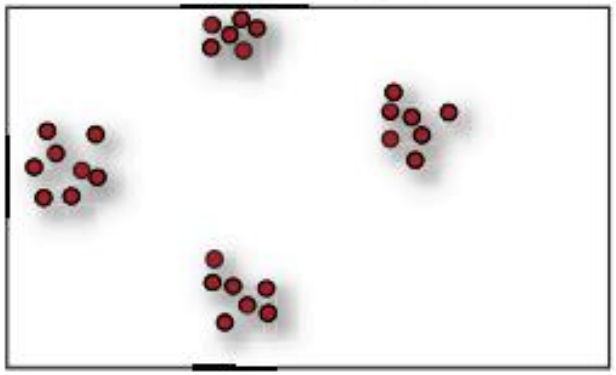
a. Random sampling.



c. Stratified random sampling.



e. Cluster sampling.



Confusion/Error Matrices

- **Confusion/error matrices (site specific)**
  - **Accepted as standards for evaluating and reporting the accuracy of classification results**
  - **Are also known as Error or Confusion or Contingency matrices because they identify the errors as well as the misclassifications and confusion among different classes**

Accuracy Assessment-Error Matrix

- **Comparison is done by creating an error matrix from which different accuracy measures can be calculated**
- **Error matrix compares on class-by-class basis the relationship between known reference data (ground truth) and corresponding results of automated classification**
- **Matrix is square, no of rows equal to number of classes**

Error Matrix

Point number	Classified Image	Reference
1	1	1
2	1	1
3	1	1
4	1	2
5	1	1
6	1	1
7	1	1
8	2	2
9	2	1
10	2	2
11	2	1
12	2	1
13	2	2
14	2	2
15	2	1
16	2	2
17	2	2
18	2	2
19	2	2
20	2	2

1. **Non Forest (10)**
2. **Forest (10)**

	Reference		
Map	NF	F	Row total
NF	6	1	7
F	4	9	13
Column total	10	10	20

Error Matrix

Classification Data	Reference Data			
	Forest	Water	Urban	Row Total
Forest	28	14	15	51
Water	1	15	5	21
Urban	1	1	20	22
Column Total	30	30	40	100

- Pixels that are correctly classified are located along the major diagonal of the error matrix
- All non diagonal elements of matrix represent error of Omission or Commission

Accuracy Measures

- **Overall classification accuracy,**
- **Producer’s accuracy (Errors of Omission)**
- **User’s (Consumers) accuracy (Errors of Commission)**

Accuracy Measures

- **The percentage of correctly classified pixels and is computed as:**

$$\textbf{Overall Accuracy (OA)} = \frac{\sum A}{\sum B} \times 100$$

  - **where, $\sum A$ is the sum of pixels assigned to correct classes (diagonal elements), and $\sum B$ is the sum of the total elements of the error matrix**

Error Matrix

- **Error of Omission and Commission**
  - **Assignment of an actual class on ground to different class on classified map**
    - **Pixels of actual class grass on ground, classified as class trees**

      is error of omission for Class Grass

      and

      error of Commission for Class Trees

Producer's Accuracy

- **Is a measure of <u>Errors of Omission</u>**
- **Is an evaluation from an analyst's point of view to see how well he was able to correctly classify ground features**
- **Informs the analyst that, e.g. of the actual forest, ---% was correctly classified.**
- $$\textbf{Producer's Accuracy(PA)} = \frac{C}{\sum C} \times 100$$
- **where *C* is the number of reference pixels correctly assigned to a class,**
- $\sum C$ **is the total number of pixels of that class from the reference data, (i.e. the respective class column total)**

User's Accuracy

- **Is the measure of the <u>Errors of Commission</u>**
- **Is the reliability of the classified data from the user's perspective and indicates the probability that pixels classified on a map represent the same class on the ground**
- **It tells the user of the map that, of the area labeled forest on the map, --% actually corresponds to forest on the ground**

$$\textbf{User Accuracy (UA)} = \frac{C}{\sum R} \times 100$$

- **where *C* is the number of reference pixels correctly assigned to a class,**
- **$\sum R$ is the total number of pixels classified to that class, (class row total)**
- **Informs the analyst that, e.g. of the actual forest, ---% was correctly classified.**

Error Matrix

(Classified Image)

Point number	Map 1	Reference
1	1	1
2	1	1
3	1	1
4	1	2
5	1	1
6	1	1
7	1	1
8	2	2
9	2	1
10	2	2
11	2	1
12	2	1
13	2	2
14	2	2
15	2	1
16	2	2
17	2	2
18	2	2
19	2	2
20	2	2

1. Non Forest
2. Forest

User's accuracy

NF= 6/7 =88 %

W = 9/13 = 69%

Producer's accuracy

NF = 6/10 = 60 %

F= 4/10 = 40 %

Overall accuracy

6+9 =15 /20= 75 %

	Reference		
Map	NF	F	Row total
NF	6	1	7
F	4	9	13
Column total	10	10	20

Error matrix of Three classes

Classification Data	F	W	U	Row total
F	**28**	**14**	**15**	**57**
W	**1**	**15**	**5**	**21**
U	**1**	**1**	**20**	**22**
Column Total	**30**	**30**	**40**	**100**

Producer's accuracy

F=28/30= 93 %

W= 15/30 =50 %

U=20/40 = 50 %

Overall accuracy

28 +15 +20 =63/100= 63 %

User's accuracy
F= 28/57 =49 %
W = 15/21 = 71%
U = 20/22 = 91 %

Kappa

- Another measure of statistical classification accuracy can be performed using the kappa (κ hat) index of agreement

$$k = \frac{N \sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})}{N^2 - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})}$$

- r: number of land cover classes (rows)
- N: total number of test pixels included in the error matrix
- $\sum x_{ii}$ total of diagonal elements
- x_{i+} : total in row i
- x_{+i}: total in column i of the matrix elements
- Kappa values are typically between 0 and 1, an indication that the observed classification is that much percent better than the results from chance assignment
- Considers the possibility of the agreement occurring by chance

Classification Data	W (water)	S (sand)	F (forest)	U (Urban)	C (corn)	H (Hay)	Row total
W	480	0	5	0	0	0	485
S	0	52	0	20	0	0	72
F	0	0	313	40	0	0	353
U	0	16	0	126	0	0	142
C	0	0	0	38	342	79	459
H	0	0	38	24	60	359	481
Column Total	480	68	356	248	402	438	1992

$\sum_{i=1}^{r} x_{ii} =$ 480+52+313+126+342+359 =1672

$\sum_{i=1}^{r} (x_{i+} \cdot x_{+i}) =$ (485*480)+(72*68)+(353*356)+(142*248)+(459*402)+(481*438) =793776

N= 1992

$\hat{K} =$ $\frac{1992(1672)-793776}{(1992)^2 -793776}$ =0.799

$$k = \frac{N \sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})}{N^2 - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})}$$

